AstroGrass: a curated resource for the Brachypodium distachyon ISS transcriptome study OSD-375 and gravitropism candidate genes

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-- DRAFT MANUSCRIPT --

Abstract

Few plant spaceflight experiments have used monocots, although monocots include most of the crops grown on Earth[Su2023]. NASA Open Science Data Repository (OSDR) study OSD-375 profiled the roots and shoots of three Brachypodium distachyon accessions (Bd21, Bd21-3 and Gaz8) grown on the International Space Station, with ground controls run at Kennedy Space Center[Su2023]. This draft describes AstroGrass, a curated resource built around that study. It contains (1) the OSD-375 sample design and growth conditions, taken from the OSDR metadata and the primary paper; (2) a catalogue of five grass spaceflight and related omics studies, each checked against its OSDR, GEO or PRIDE record; (3) a list of 29 gravitropism candidate genes defined by Arabidopsis anchor genes, of which 17 anchor identifiers were confirmed in UniProt and 12 were found to name other genes and are marked for correction; and (4) time-lapse videos of Brachypodium root gravitropism. No new analysis of the OSD-375 data is reported. Mapping the candidates to Brachypodium genes, extracting their spaceflight expression, measuring gravitropism across accessions and comparing with Arabidopsis are planned and are listed as such.

Keywords: *Brachypodium distachyon, gravitropism, spaceflight, microgravity, NASA OSDR, OSD-375, APEX-06, ISS, transcriptomics, monocot, data resource*

Introduction

Plants sense the direction of gravity mainly through the movement of starch-filled plastids (amyloplasts) that act as statoliths in specialised gravity-sensing cells (statocytes)[Morita2010]. In Arabidopsis, LAZY1-LIKE (LZY) proteins associate with amyloplasts and the adjacent plasma membrane and become polarised according to the direction of gravity[Nishimura2023,Chen2023]. At the plasma membrane, LZY recruits RCC1-like domain (RLD) proteins, which are required for polar auxin transport[Furutani2020], and in the root columella the auxin efflux regulator PIN3 relocalises laterally after gravity stimulation[Friml2002].

In rice, root growth angle is controlled in part by DEEPER ROOTING 1 (DRO1) and its homolog qSOR1[Uga2013,Kitomi2020]. Crops such as wheat have been studied for decades as components of life support systems for space[Wheeler2017], yet few plant spaceflight experiments have used monocots[Su2023].

Brachypodium distachyon is a model for temperate grasses. It was the first member of the Pooideae, the grass subfamily that includes wheat and barley, to have its genome sequenced[IBI2010], and de novo assemblies of 54 lines define its pan-genome[Gordon2017].

In the APEX-06 experiment (NASA OSDR study OSD-375), seedlings of the Bd21, Bd21-3 and Gaz8 accessions were germinated and grown on the International Space Station (ISS) and in a ground control[Su2023]. Su et al. reported that the roots of all three accessions grew more slowly and had longer root hairs in microgravity, and that the number of differentially expressed genes (DEGs) differed widely between accessions: 4,527 in Gaz8, 1,353 in Bd21 and 570 in

Bd21-3, with only 47 shoot DEGs and six root DEGs shared between accessions[Su2023].

This draft describes the AstroGrass resource as it now stands: what is in it, where each item comes from, and what remains to be done.

Status of this draft

Versions of this manuscript and its website before 14 October 2026 reported a 15-accession gravitropism survey, a genome-wide association study (GWAS), a re-analysis of OSD-375 with gene-level fold changes, a cross-species comparison with Arabidopsis, and synteny, promoter-motif and alternative-splicing analyses. These results have been withdrawn. The gravitropism measurements had been generated by the pipeline's demonstration mode from seeded random numbers, no genotype data or association test existed, and the remaining values were typed into the code rather than computed. The Brachypodium gene identifiers given for the candidate genes were also withdrawn, because none could be confirmed as an ortholog of its Arabidopsis anchor. The sections below report only what the repository now contains.

Results

OSD-375 study design

OSD-375 contains 48 RNA-seq samples: three accessions, two organs (roots and shoots), flight or ground control, and four APEX Growth Units per accession and condition (Figure 1). Each of the 24 growth units gave one root and one shoot sample. Seedlings were germinated for 24 h under red light and then grown for four days under continuous red, blue and green light at about 40-55 µmol m?^2 s?^1 and 22 degC; the flight samples received an absorbed radiation dose of 1.397 mGy (OSDR metadata;[Su2023]). The ground control was run in a VEGGIE unit inside the ISS Environmental Simulator at Kennedy Space Center with a two-day delay[Su2023]. Supplementary Figure S1 summarises the hardware, timeline and conditions.

All 48 libraries were sequenced on Illumina NovaSeq 6000 instruments, but Bd21 was sequenced at the University of Wisconsin-Madison with 2 x 150 bp reads, and Bd21-3 and Gaz8 by the NASA GeneLab Sample Processing Lab with 2 x 100 bp reads[Su2023]. Accession is therefore confounded with sequencing batch. Flight-versus-ground comparisons within an accession are not affected, but differences between accessions in the number of DEGs cannot be separated from batch effects.

Study catalogue

AstroGrass catalogues five studies (Figure 2): OSD-375[Su2023]; OSD-622, a microarray study of leaves of dwarf wheat (cultivar USU Apogee) grown on the ISS in the Biomass Production System during the PESTO experiment[Stutte2006]; two ground RNA-seq studies of Brachypodium, GSE97940 (roots treated with 2,4-D to induce nodule-like structures)[Thomas2018] and GSE48040 (cold-treated seedlings of accession ABR5)[Zhang2013]; and the leaf phosphoproteome PXD000868[Lv2014]. Each entry was checked against its OSDR, GEO or PRIDE record, and each DOI against CrossRef or DataCite. Fields that a record does not state are marked as such rather than filled in.

Gravitropism candidate genes

The candidate list contains 29 genes chosen from the Arabidopsis gravitropism literature: PIN and AUX/LAX auxin carriers, LAZY/DRO and SGR genes, TIR1/AFB auxin receptors, ARF and Aux/IAA transcription factors, starch-synthesis genes, calcium and mechanosensing genes, and cell-wall genes. Each candidate is defined by an Arabidopsis anchor gene. For 17 candidates the Arabidopsis identifier (AGI) given in the list was confirmed: UniProt lists

the intended gene name for that AGI. For the other 12, the AGI names a different gene; for example, the AGI given for LAZY1 is PDF1B, the AGI given for SGR9 is LAZY2, and the AGI given for MSL10 is MSL1. Those 12 anchors are marked TODO in the table, together with the gene each wrong AGI actually names.

No Brachypodium gene identifiers are assigned yet. The earlier list gave a BRADI identifier for each candidate. Nine of the 29 do not exist in Ensembl Plants (Ensembl Genomes release 63), and for the 17 candidates with a verified anchor, none of the given identifiers is among the anchor's Brachypodium orthologs in Ensembl Compara (data/genotypes/withdrawn_bradi_id_check.csv). All 29 were therefore withdrawn. Two markers in the earlier table, a serine hydroxymethyltransferase and the chloroplast-encoded RuBisCO large subunit, were not gravitropism candidates and were dropped with the typed-in expression values attached to them.

Time-lapse videos and kinetics plots

The website hosts time-lapse videos of Brachypodium roots of accessions Per1 and BdTR7a after a 90 deg rotation, from the author's CyVerse Data Store. The repository also holds per-accession plots of root orientation and root-tip displacement over time for 45 accessions, added to the repository as Barker 2016 data. The numbers behind these plots are not in the repository, so no statistics are drawn from them here.

Planned analyses (not yet run)

TODO: Correct the 12 Arabidopsis anchors, then map Brachypodium orthologs of all 29 candidates with Ensembl Compara, recording the Ensembl release and any one-to-many cases.

TODO: Extract the expression of the candidate genes from the per-accession DEG tables published by Su et al.[Su2023] (supplementary tables, CC BY 4.0), crediting the values to that study.

TODO: If a new differential-expression analysis of OSD-375 is needed, start from the FASTQ files (the only expression data in OSDR for this study) and account for the confounding of accession with sequencing batch.

TODO: Add gravitropism measurements for the accession panel, with their source documented. The kinetics plots above would need their underlying numbers before any comparison between accessions.

TODO: An association study would need genotype data for the phenotyped accessions; with tens of accessions it would be a candidate-gene comparison rather than a GWAS.

TODO: Compare Brachypodium spaceflight responses with the author's Arabidopsis spaceflight analysis, which is in preparation.

Methods

OSD-375 metadata

The OSD-375 file listing and ISA-Tab metadata were downloaded from the OSDR API (https://osdr.nasa.gov/osdr/data/osd/files/375) by code/01_download_osdr_osd375.py. The sample table (s_GLDS-375.txt) was parsed by code/02_parse_osdr_metadata.py to count samples per accession, organ and condition (Figure 1). Growth conditions and the radiation dose were read from the same table; the timeline and sequencing details are from Su et al.[Su2023].

Study catalogue

Catalogue entries (code/10_build_astrograss_db.py) were checked on 14 October 2026 against the OSDR study

metadata API, NCBI GEO (E-utilities and the GEO sample records), the PRIDE Archive API, CrossRef and DataCite.

Candidate-gene anchors

For each AGI in the curated list (data/genotypes/candidate_genes_curated.csv), code/14_check_arabidopsis_anchors.py retrieved the reviewed UniProt entry cross-referenced to that Araport identifier (UniProt release 2026_03). An anchor was accepted only if the intended Arabidopsis gene symbol was among the entry's gene names. The same script looked up each withdrawn BRADI identifier in Ensembl (REST API, Ensembl Genomes release 63) and, for verified anchors, retrieved the anchor's Brachypodium orthologs from Ensembl Compara (plants). code/03_candidate_genes.py combines the list and the anchor check into data/genotypes/gravitropism_candidate_genes.csv.

Data and code availability

All code and tables are available at https://github.com/dr-richard-barker/brachypodium-gwas-spaceflight, and the website at https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/. Raw sequencing data are available from NASA OSDR under accession OSD-375 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-375; DOI: 10.26030/2x6b-3v89).

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Author Contributions

R.B. conceived the study, curated the resource, built the data pipeline and website, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

References

[Su2023] Su SH, Levine HG, Masson PH (2023). Brachypodium distachyon Seedlings Display Accession-Specific Morphological and Transcriptomic Responses to the Microgravity Environment of the International Space Station. Life 13(3):626. doi:10.3390/life13030626

[Morita2010] Morita MT (2010). Directional Gravity Sensing in Gravitropism. Annual Review of Plant Biology 61(1):705-720. doi:10.1146/annurev.arplant.043008.092042

[Nishimura2023] Nishimura T, Mori S, Shikata H, et al. (2023). Cell polarity linked to gravity sensing is generated by LZY translocation from statoliths to the plasma membrane. Science 381(6661):1006-1010. doi:10.1126/science.adh9978

[Chen2023] Chen J, Yu R, Li N, et al. (2023). Amyloplast sedimentation repolarizes LAZYs to achieve gravity sensing in plants. Cell 186(22):4788-4802.e15. doi:10.1016/j.cell.2023.09.014

[Furutani2020] Furutani M, Hirano Y, Nishimura T, et al. (2020). Polar recruitment of RLD by LAZY1-like protein during gravity signaling in root branch angle control. Nature Communications 11(1):76. doi:10.1038/s41467-019-13729-7

[Friml2002] Friml J, Wi?niewska J, Benková E, Mendgen K, Palme K (2002). Lateral relocation of auxin efflux regulator PIN3 mediates tropism in Arabidopsis. Nature 415(6873):806-809. doi:10.1038/415806a

[Uga2013] Uga Y, Sugimoto K, Ogawa S, et al. (2013). Control of root system architecture by DEEPER ROOTING 1 increases rice yield under drought conditions. Nature Genetics 45(9):1097-1102. doi:10.1038/ng.2725

[Kitomi2020] Kitomi Y, Hanzawa E, Kuya N, et al. (2020). Root angle modifications by the DRO1 homolog improve rice yields in saline paddy fields. Proceedings of the National Academy of Sciences 117(35):21242-21250. doi:10.1073/pnas.2005911117

[Wheeler2017] Wheeler RM (2017). Agriculture for Space: People and Places Paving the Way. Open Agriculture 2(1):14-32. doi:10.1515/opag-2017-0002

[IBI2010] The International Brachypodium Initiative (2010). Genome sequencing and analysis of the model grass Brachypodium distachyon. Nature 463(7282):763-768. doi:10.1038/nature08747

[Gordon2017] Gordon SP, Contreras-Moreira B, Woods DP, et al. (2017). Extensive gene content variation in the Brachypodium distachyon pan-genome correlates with population structure. Nature Communications 8(1):2184. doi:10.1038/s41467-017-02292-8

[Stutte2006] Stutte GW, Monje O, Hatfield RD, Paul AL, Ferl RJ, Simone CG (2006). Microgravity effects on leaf morphology, cell structure, carbon metabolism and mRNA expression of dwarf wheat. Planta 224(5):1038-1049. doi:10.1007/s00425-006-0290-4

[Thomas2018] Thomas J, Bowman MJ, Vega A, Kim HR, Mukherjee A (2018). Comparative transcriptome analysis provides key insights into gene expression pattern during the formation of nodule-like structures in Brachypodium. Functional & Integrative Genomics 18(3):315-326. doi:10.1007/s10142-018-0594-z

[Zhang2013] Zhang J, Mao Z, Chong K (2013). A global profiling of uncapped mRNAs under cold stress reveals specific decay patterns and endonucleolytic cleavages in Brachypodium distachyon. Genome Biology 14(8):R92. doi:10.1186/gb-2013-14-8-r92

[Lv2014] Lv DW, Li X, Zhang M, et al. (2014). Large-scale phosphoproteome analysis in seedling leaves of Brachypodium distachyon L. BMC Genomics 15(1):375. doi:10.1186/1471-2164-15-375